Learning History-Dependent Constitutive Laws From Data

Mayank Raj

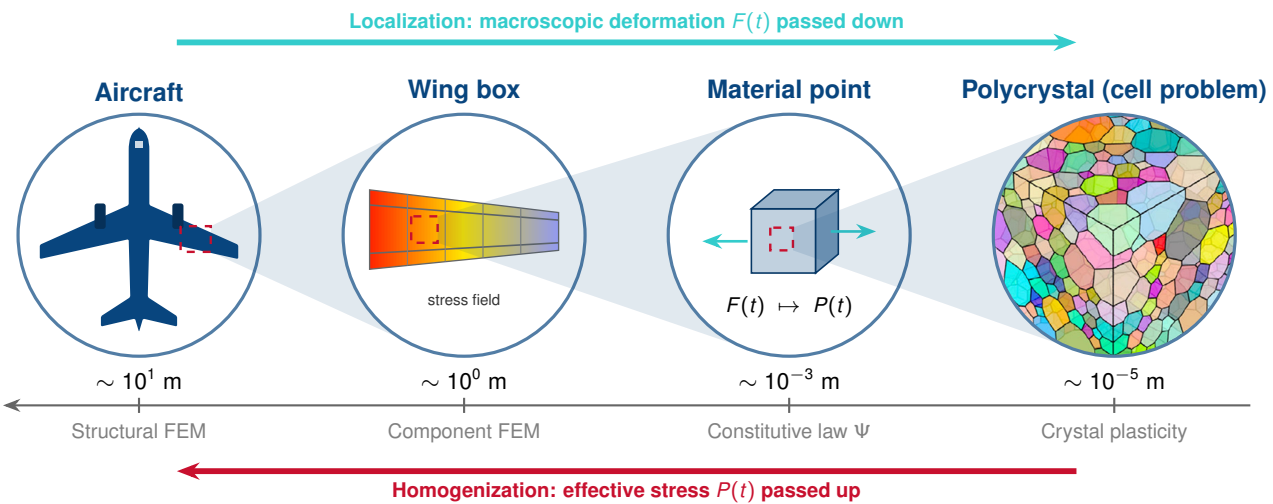
Lianghao Cao, Andrew Stuart, Kaushik Bhattacharya

MURI Meeting Talk

October 22, 2026

Raj, M., Cao, L., Stuart, A., & Bhattacharya, K. (2026). A neural-network framework to learn history-dependent constitutive laws and identifiability of internal variables. *Journal of the Mechanics and Physics of Solids*, *217*, 106807

Multiscale Modelling: From Microstructure to Aircraft



Why not resolve every grain?

Wing $\sim$ 10 m vs. grains $\sim 10\,\mu$m $\implies$ 10^{6} per direction, $\sim 10^{18}$ cells in 3D, tracked over the full load history.

Learn a (fast) surrogate for stress-strain (P-F) constitutive law, bypassing (slow) polycrystal simulations.

Modelling Plasticity: Traditional vs. Data-Driven

Traditional: Phenomenological Plasticity

$$\mathbf{F} = \mathbf{F}^e\mathbf{F}^p, \qquad \mathbf{P} = \partial_{\mathbf{F}}\psi(\mathbf{F}\mathbf{F}^{p-1}, q),$$
$$\phi(\boldsymbol{\Sigma}, q) \leq 0 \ \text{(yield surface)}, \qquad \boldsymbol{\Sigma} = \mathbf{F}^{e\top}\partial_{\mathbf{F}^e}\psi,$$
$$\dot{\mathbf{F}}^p\mathbf{F}^{p-1} = \dot{\lambda}\,\partial_{\boldsymbol{\Sigma}}\phi, \qquad \dot{q} = \dot{\lambda}\,h(\boldsymbol{\Sigma}, q).$$

- ✓ Thermodynamically consistent by construction
- ✗ Phenomenological macroscale model, may not capture all microstructural effects
- ✗ Computationally expensive in multiscale simulations

Data-Driven: Recurrent Neural Operator

$$P(t) = f\big(F(t), \dot{F}(t), \xi(t)\big),$$
$$\dot{\xi}(t) = g\big(F(t), \xi(t)\big), \qquad \xi(0) = 0.$$

f, g: neural networks; ξ: learned internal variables

- ✓ Learned from cell-problem (crystal plasticity) data
- ✓ Fast and accurate surrogate for the macro-scale solver
- ✗ Not thermodynamically consistent in general

This work

- Constrain f, g in terms of energy density $\mathcal{W}$ and dissipation potential $\mathcal{D}$.
- Leads to thermodynamic consistency.
- Allows imposition of well-posedness on macro-problem.

Bhattacharya et al., 2023; Liu et al., 2023; Simo and Hughes, 1998

Inspiration: Energetic Formulation of Material Models

Ball (1976, 1977)

Nonlinear elasticity: polyconvexity and coercivity of the energy

$$\mathcal{W}(\mathbf{F}) = g(\mathbf{F}, \operatorname{adj}\mathbf{F}, \det\mathbf{F}), \quad g \text{ convex}$$

- existence of energy minimizers (solutions of the macro problem).

Ortiz & Stainier (1999)

Variational constitutive updates: internal variables minimize an incremental potential

$$\xi_{n+1} = \arg\min_{\xi} \Big[\mathcal{W}(\mathbf{F}_{n+1}, \xi) + \Delta t\, \mathcal{D}\big(\tfrac{\xi - \xi_n}{\Delta t}\big)\Big]$$

- material model fully defined by $\mathcal{W}$ and $\mathcal{D}$.

Mielke (2003, 2004, 2018)

Energetic formulation of finite-strain elasto-plasticity: stored energy $\mathcal{E}$ and dissipation distance $\mathcal{D}_{\text{dist}}$

$$\mathcal{E}(t, q(t)) \le \mathcal{E}(t, \tilde{q}) + \mathcal{D}_{\text{dist}}(q(t), \tilde{q})$$

- existence of solutions for (visco)plasticity.

Our Data-Driven Approach: Learn $\mathcal{W}$ and $\mathcal{D}$ From Data

$$\mathbf{P}(t) = \partial_{\mathbf{F}}\mathcal{W}(\mathbf{F}(t), \xi(t)), \qquad \partial_{\dot{\xi}}\mathcal{D}(\dot{\xi}(t)) + \partial_{\xi}\mathcal{W}(\mathbf{F}(t), \xi(t)) = 0, \qquad \xi(0) = 0.$$

- Thermodynamically consistent: Energy and convex dissipation potentials exist by construction
- Existence of a minimizer: Polyconvexity and coercivity of $\mathcal{W}$, and convexity of $\mathcal{D}$ are built into the architecture

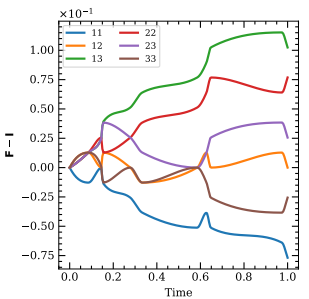
Ball, 1977; Ball, 1976; Mielke, 2003; Mielke, 2004; Mielke et al., 2018; Ortiz and Stainier, 1999

Part I

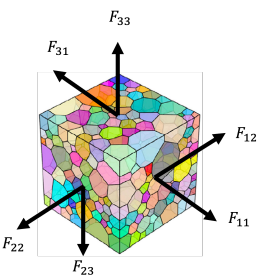
Learning for fixed Microstructure

Crystal Plasticity Data

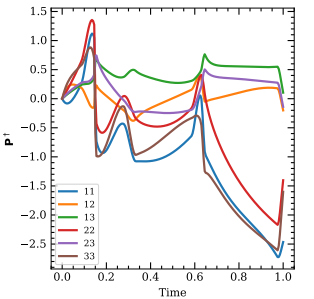
Input: deformation $F(t)$



Polycrystal unit cell



Output: stress $P(t)$



Offline high-fidelity crystal plasticity simulations of the polycrystal unit cell provide trajectories of deformation and stress:

Data

$$\{F^{(i)}(t), P^{(i)}(t)\}_{i=1}^{N}$$

Liu et al. (2023) approximate the crystal plasticity boundary value problem using Taylor averaging.

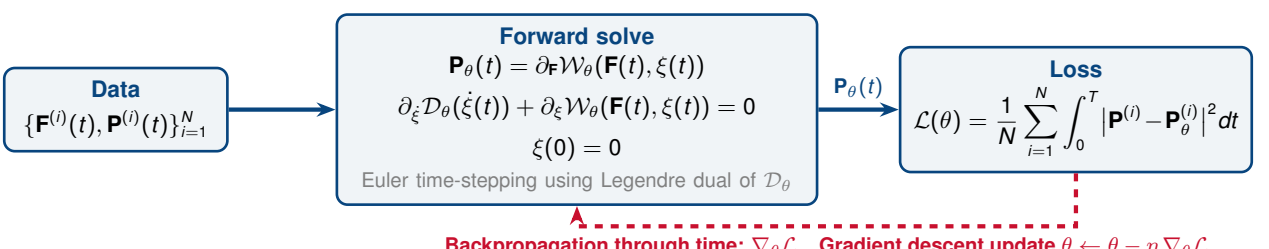
Liu et al., 2023

Parameterization using Neural Network

Parameterization of $\mathcal{W}$ and $\mathcal{D}$

$$\mathcal{W}_\theta(\mathbf{F}, \xi) = \mathcal{N}_H(\det \mathbf{F}) + \mathcal{N}_V(\mathbf{F}, \operatorname{adj} \mathbf{F}, \det \mathbf{F}, \xi) + \mathcal{G}_1(\det \mathbf{F}) + \textstyle\sum_{ij} \mathcal{G}_2(\mathbf{F}_{ij}) + \mathcal{G}_2(\det \mathbf{F}),$$
$$\mathcal{D}_\theta(\dot{\xi}) = \mathcal{N}_D(\dot{\xi}) + \mathcal{G}_2(\dot{\xi}).$$

$\mathcal{N}_H, \mathcal{N}_V, \mathcal{N}_D$: convex by architecture design; $\mathcal{G}_1, \mathcal{G}_2$: coercivity of $\mathcal{W}$ and $\mathcal{D}$.



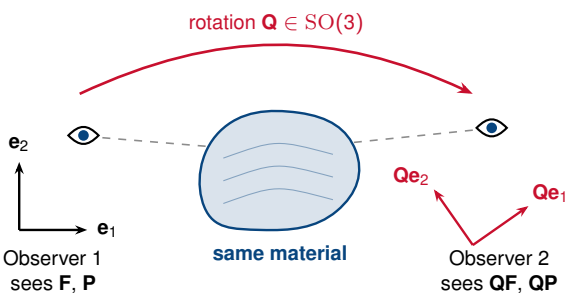
Internal Variables Are Latent

ξ is neither measured nor supervised: it never appears in the dataset or the loss. It is discovered implicitly, only through how well the predicted stress $\mathbf{P}_\theta$ matches the data.

Frame Indifference

Definition (Requirement)

$$\mathcal{W}(\mathbf{QF}) = \mathcal{W}(\mathbf{F})$$
$$\mathbf{P}(\mathbf{QF}) = \mathbf{QP}(\mathbf{F}), \quad \mathbf{Q} \in \mathrm{SO}(3)$$



both observers measure the same stored energy $\mathcal{W}$

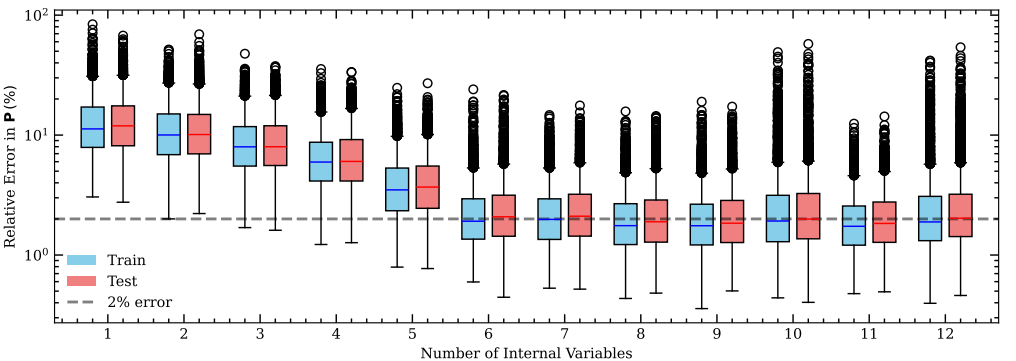
Extension of the Surrogate Model that is Consistent With Frame Indifference

$$\mathbf{P}(\mathbf{F}) := \begin{cases} \mathbf{P}_D(\mathbf{F}), & \text{if } \mathbf{F} \in \mathbb{R}^{3\times3}_{sym}; \\ \widetilde{\mathbf{Q}}\mathbf{P}_D(\widetilde{\mathbf{F}}), & \text{if } \mathbf{F} \neq \mathbb{R}^{3\times3}_{sym},\ \mathbf{F} = \widetilde{\mathbf{Q}}\widetilde{\mathbf{F}},\ \widetilde{\mathbf{Q}} \in \mathrm{SO}(3),\ \widetilde{\mathbf{F}} \in \mathbb{R}^{3\times3}_{sym}. \end{cases}$$

where $\mathbf{P}_D$ is the stress map trained on data.

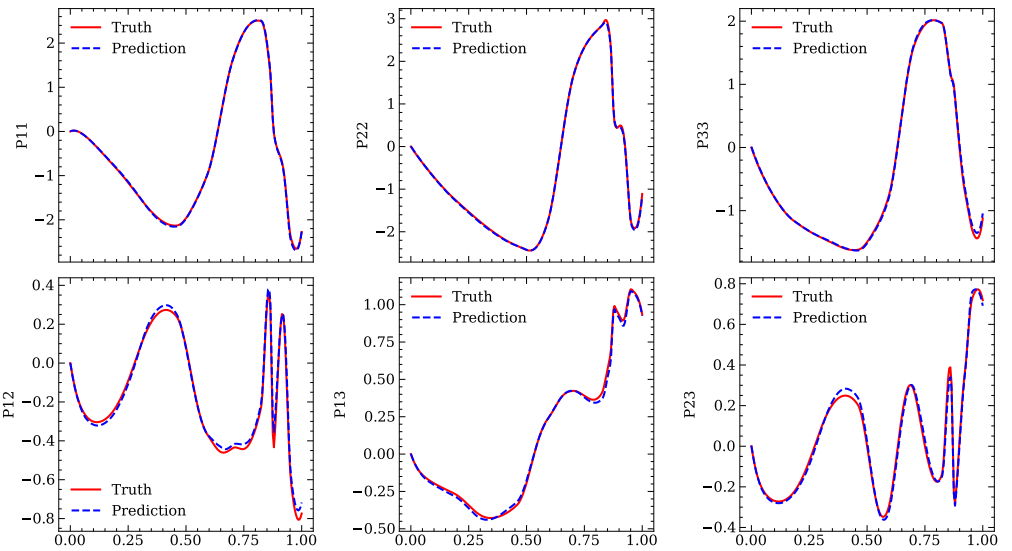
Optimal number of Internal Variables

The number of internal variables are unknown apriori!



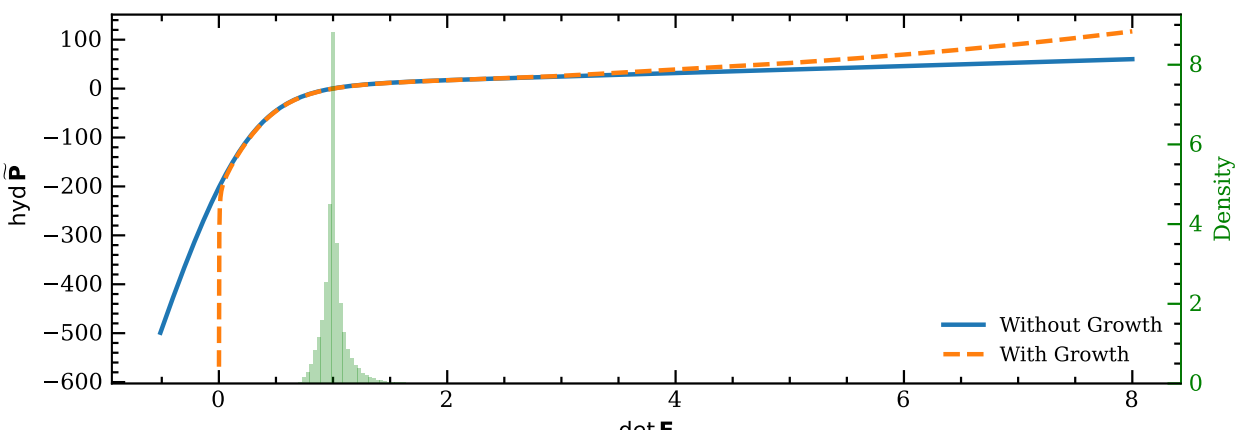
Six is the optimal number of internal variables in this example.

Comparison of True and Predicted Stress



COERCIVITY

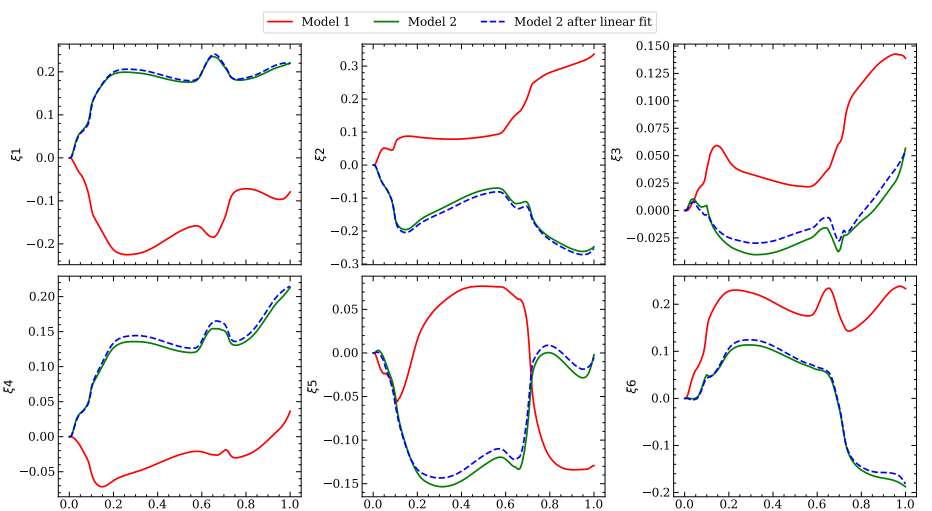
$$\mathcal{W}(\mathbf{F},\xi) = \mathcal{N}_H(\det \mathbf{F}) + \mathcal{N}_V(\mathbf{F}, \operatorname{adj}\mathbf{F}, \det \mathbf{F}, \xi) + +\mathcal{G}_1(\det \mathbf{F}) + \sum_{ij} \mathcal{G}_2(\mathbf{F}_{ij}) + \mathcal{G}_2(\det \mathbf{F});$$



Comparison of Internal Variables

Different initializations of training lead to different internal variables, ξ and η. How are they related?

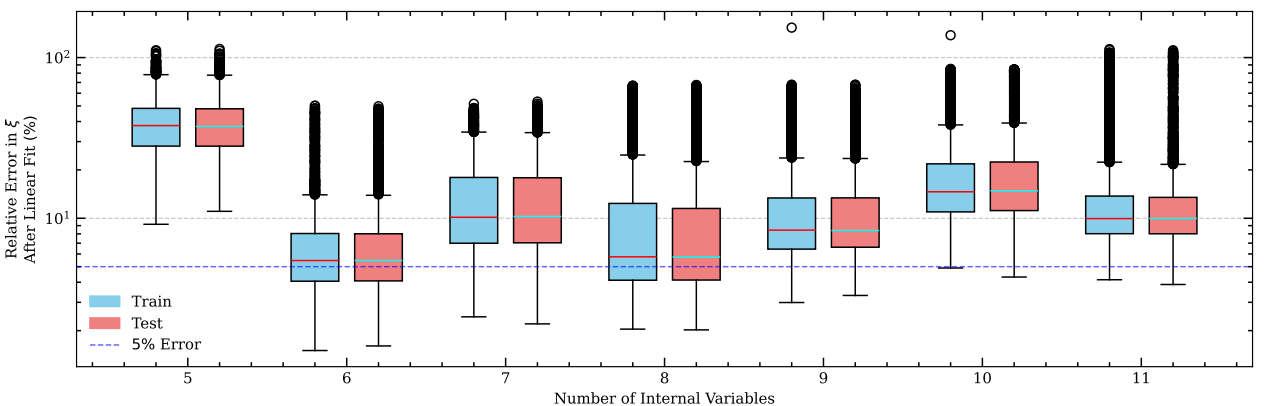
$$A^\star = \underset{A \in \mathbb{R}^{n \times n}}{\arg\min} \sum_{i=1}^{N} \int_0^T \left| \xi^{(i)}(t) - A\,\eta^{(i)}(t) \right|^2 dt$$



Error in the Linear Fit of Internal Variables

$$A^\star = \underset{A\in\mathbb{R}^{n\times n}}{\arg\min} \sum_{i=1}^{N} \int_0^T \left|\xi^{(i)}(t) - A\,\eta^{(i)}(t)\right|^2 dt$$

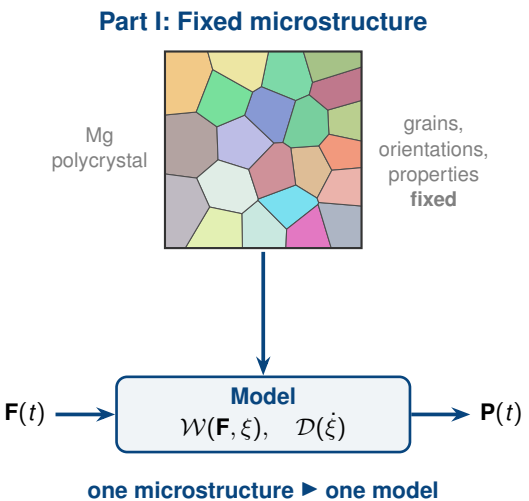
$$e^{(i)} = \frac{\left\|\xi^{(i)} - A^\star \eta^{(i)}\right\|_{L^2(0,T)}}{\left\|\xi^{(i)}\right\|_{L^2(0,T)}}$$

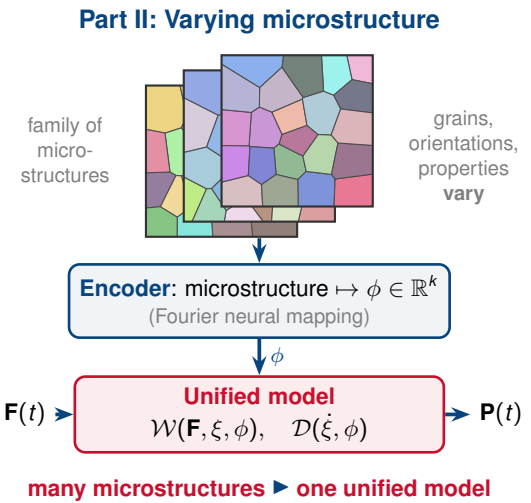


Part II

Microstructure-dependent Learning

From a Fixed Microstructure to a Family of Microstructures





Part I

Data: $\{\mathbf{F}^{(i)}, \mathbf{P}^{(i)}\}_{i=1}^{N}$ for a single microstructure. A new microstructure requires new data and retraining.

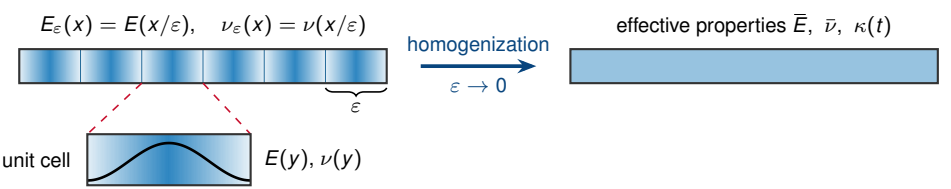
Part II

Data: $\{\text{microstructure}^{(i)}, \mathbf{F}^{(i)}, \mathbf{P}^{(i)}\}_{i=1}^{N}$. One model predicts the response of unseen microstructures.

Multiscale Kelvin-Voigt Model: Homogenization

Kelvin-Voigt Viscoelasticity Model (Multiscale Problem)

$$\nabla_x \cdot \left(E_\varepsilon : \partial_x u_\varepsilon + \nu_\varepsilon : \partial_{tu}^2 u_\varepsilon \right) = 0, \quad (x,t) \in \Omega \times \mathcal{T}, \text{B.C and I.C.}$$



Homogenized Kelvin-Voigt Viscoelastic Model (Macro Problem)

$$P^\dagger = \overline{E} : \partial_x u_0 + \bar{\nu} : \partial_{tu}^2 u_0 - \int_0^t \kappa(t-s) : \partial_x u_0(s)\, ds,$$

Multiscale Kelvin-Voigt Model: Internal Variables

Markovian Approximation of the Homogenized KV Model

$$P_0^\dagger = \overline{E} : F + \overline{\nu} : \dot{F} - \sum_{l=1}^{L} \xi_l,$$
$$\partial_t \xi_l = \beta_l F - \alpha_l \xi_l, \quad \xi_l(0) = 0, \quad l \in \{1, \dots, L\},$$

Energy Density and Dissipation

$$\mathcal{W}^\dagger(F, \{\xi_l\}_{l=1}^L) = \frac{1}{2}\overline{E}F^2 - F\sum_{l=1}^{L}\xi_l + \sum_{l=1}^{L}\frac{\alpha_l}{2\beta_l}\xi_l^2$$
$$\mathcal{D}^\dagger(\dot{F}, \{\dot{\xi}_l\}_{l=1}^L) = \frac{1}{2}\overline{\nu}\dot{F}^2 + \sum_{l=1}^{L}\frac{1}{2}\frac{\dot{\xi}_l^2}{\beta_l};$$

Two Forms of the Homogenized Kelvin-Voigt Model

Form 1: Convolution With the Strain

$$P^\dagger = \overline{E}F + \bar{\nu}\dot{F} - \int_0^t \kappa(t-s)\,F(s)\,ds$$

$$\overline{E} = \frac{\langle E/\nu^2\rangle}{\langle 1/\nu\rangle^2}, \qquad \bar{\nu} = \langle 1/\nu\rangle^{-1}, \qquad \langle\cdot\rangle : \text{cell average}$$

Bhattacharya et al., 2023; Sanchez-Palencia, 1980

Form 2: Convolution With the Strain Rate

$$P^\dagger = \widetilde{E}F + \bar{\nu}\dot{F} + \int_0^t K(t-s)\,\dot{F}(s)\,ds$$

$$\widetilde{E} = \langle 1/E\rangle^{-1}, \qquad \bar{\nu} = \langle 1/\nu\rangle^{-1} \text{ (same viscosity)}$$

Francfort and Suquet, 1986

Related Through Integration by Parts

With $K(\tau) = \int_\tau^\infty \kappa(r)\,dr$ and $F(0) = 0$,

$$\int_0^t \kappa(t-s)\,F(s)\,ds = K(0)\,F(t) - \int_0^t K(t-s)\,\dot{F}(s)\,ds.$$

- The same material response and the same viscosity $\bar{\nu}$, but a different elastic part: $\widetilde{E} = \overline{E} - K(0) = \langle 1/E\rangle^{-1}$ (harmonic mean).
- The homogenized elasticity is not unique: it depends on which form is chosen.

Meaning of Internal Variables

Phenomenological Plasticity



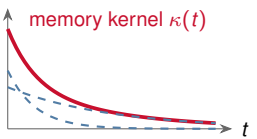
$\xi = \{\mathbf{F}^p, \lambda\}$

- $\mathbf{F}^p$: plastic deformation
- λ: measure of crystal slip and twinning

Physical: internal variables represent physical quantities.

physical meaning

Kelvin-Voigt Viscoelasticity



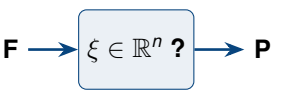
$\kappa(t) \approx \sum_{l=1}^{L} \beta_l e^{-\alpha_l t}$

- $\xi_l(t) = \int_0^t \beta_l e^{-\alpha_l (t-s)} F(s)\, ds$
- $\dot{\xi}_l = \beta_l F - \alpha_l \xi_l$

Approximation-theoretic: internal variables come from approximating the memory kernel.

mathematical approximation

Data-Driven Models



ξ learned from $(\mathbf{F}, \mathbf{P})$ data only

- never observed or supervised
- identifiable only up to a linear transform, $\xi = A\eta$

No a priori meaning: no hypothesis that ξ represents any physical quantity.

purely latent

Neural Network Architecture

Data

$$\left\{E^{(i)}, \nu^{(i)}, F^{(i)}, P^{(i)}\right\}_{i=1}^{N}$$

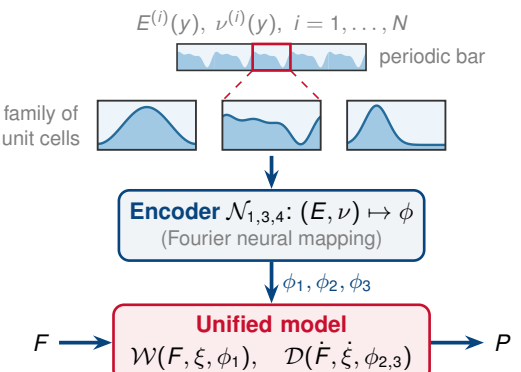
Energy Function Architecture

$$\overbrace{\phi_1}^{\text{vector}} = \mathcal{N}_1(\overbrace{E, \nu}^{\text{Functions}}),$$
$$\mathcal{W} = \mathcal{N}_2(\underbrace{F, \xi, \phi_1}_{\text{vectors}}),$$

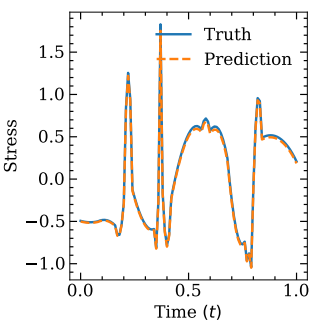
Dissipation Function Architecture

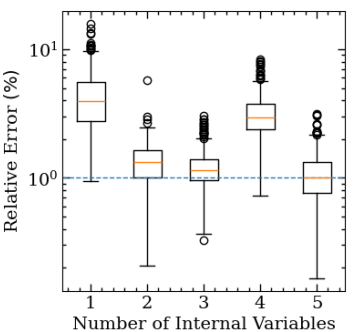
$$\phi_2 = \mathcal{N}_3(E, \nu), \quad \phi_3 = \mathcal{N}_4(E, \nu),$$
$$\mathcal{D} = -\mathcal{N}_5(\dot{F}, \phi_2) + \mathcal{N}_6(\dot{\xi}, \phi_3),$$

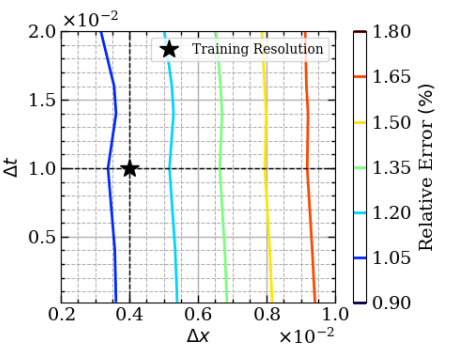
$\mathcal{N}_{1,3,4}$: Fourier Neural Mappings.
$\mathcal{N}_{2,5,6}$: Convex Architecture



Stress Prediction Accuracy

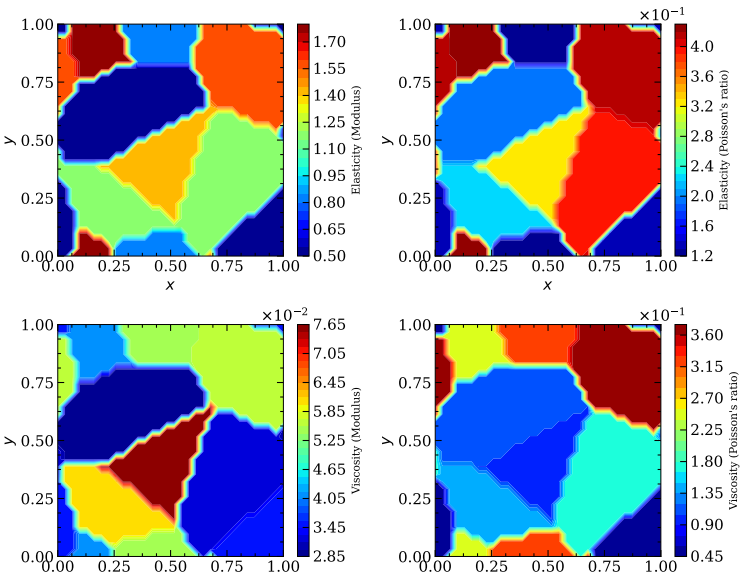




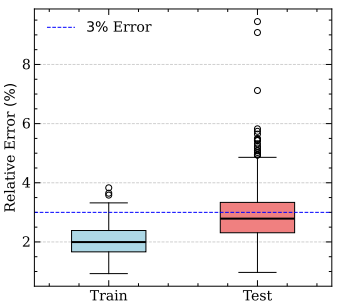


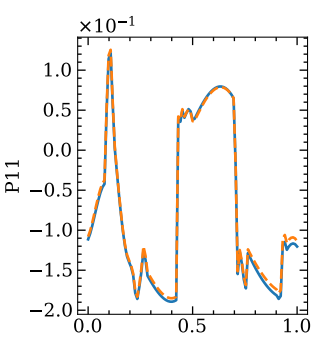
Two-Dimensional Kelvin-Voigt Multiscale Model

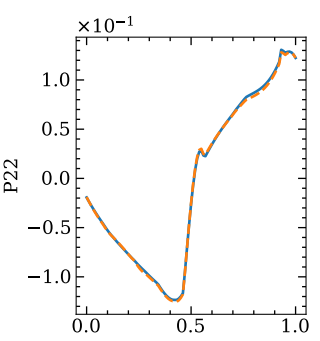
$$\nabla \cdot (E_\varepsilon : \nabla u_\varepsilon + \nu_\varepsilon : \nabla \partial_t u_\varepsilon) = 0, \quad (x, t) \in \Omega \times \mathcal{T}, \text{B.C and I.C.}$$

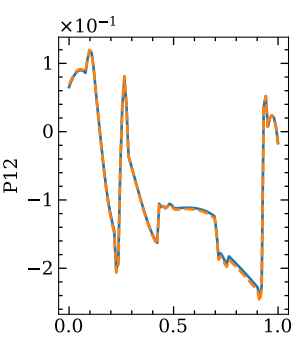


Stress Prediction Accuracy

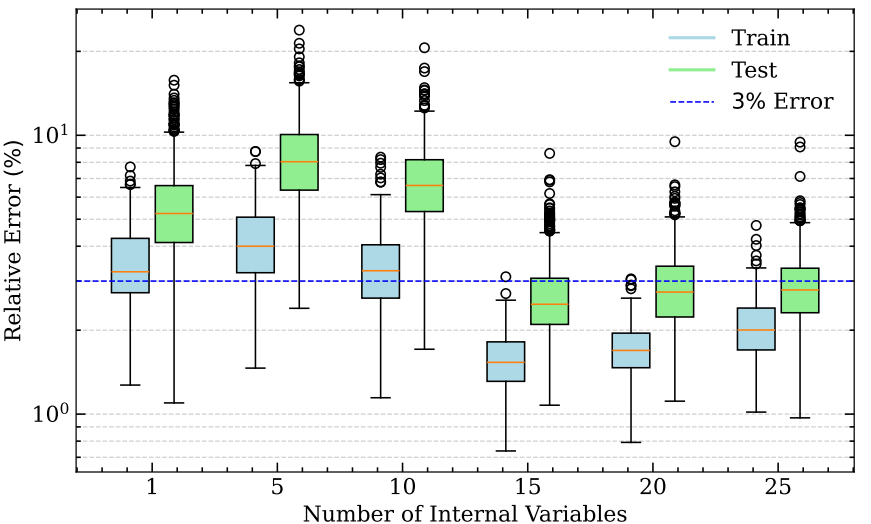








Number of Internal Variables



Non-Linear Model- Viscoplasticity

Multiscale Model

$$P_\varepsilon(t) = E_\varepsilon(x)(F_\varepsilon(t) - \xi_\varepsilon(t)),$$
$$\dot{\xi}_\varepsilon(t) = c_{0\varepsilon}(x)\mathrm{sign}(P_\varepsilon(t)) \left(\frac{|P_\varepsilon(t)|}{P_{0\varepsilon}(x)}\right)^{n_\varepsilon(x)}$$

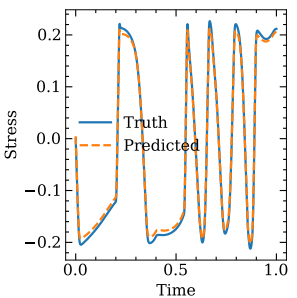
Data

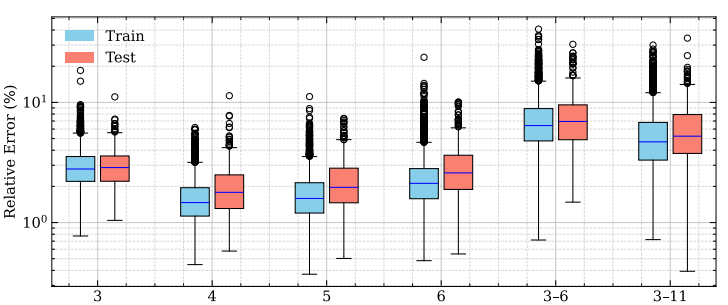
$$\left\{E^{(i)}, c_0^{(i)}, P_0^{(i)}, n^{(i)}, F^{(i)}, P^{(i)}\right\}_{i=1}^{N}$$

Homogenized Model

$$\mathcal{W}^\dagger(F,\xi) = \frac{1}{2}\bar{E}(F-\xi)^2,$$
$$\mathcal{D}^{\dagger *}(q) = \int_{\mathbb{T}} c_0(y)P_0(y)^{-n(y)} \left(\frac{1}{n(y)+1}|q|^{n(y)+1}\right) dy.$$

Predictive Accuracy





CONTRIBUTIONS

- Framework for learning surrogates that are consistent with the second law of thermodynamics.
- Framework learning surrogates that are consistent with convexity and coercivity of the energy function.
- Identifiability of internal variables up to linear transform and supporting theory.
- The framework is deployed and demonstrated to work in various settings: crystal plasticity, Kelvin-Voigt viscoelasticity and viscoplasticity.
- Famework developed to allow learning of microstructure-dependence in constitutive laws.

Raj, M., Cao, L., Stuart, A., & Bhattacharya, K. (2026). A neural-network framework to learn history-dependent constitutive laws and identifiability of internal variables. *Journal of the Mechanics and Physics of Solids*, *217*, 106807

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